The Roles of Landuse and Centrality in Urban Land Productivity: An Empirical Study in Lafayette, Louisiana

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ABSTRACT

This study explores the impact that landuse and centrality have on property values at the parcel level in Lafayette Parish, Louisiana. The spatial hedonic regression model developed serves to determine to what extent different centrality measures (distance to the central business district (CBD), population weighted distance, Betweenness and Straightness) affect land value per acre across residential, commercial and other property uses. Results indicate that all centrality measures were statistically significant for at least one landuse category. Increased centrality has a mostly positive impact on value per acre across all landuse categories when centrality measures are tested individually but shows mixed effects when centrality measures are combined into one model.

MOTIVATION

← Suburbs of Santa Barbara, CA

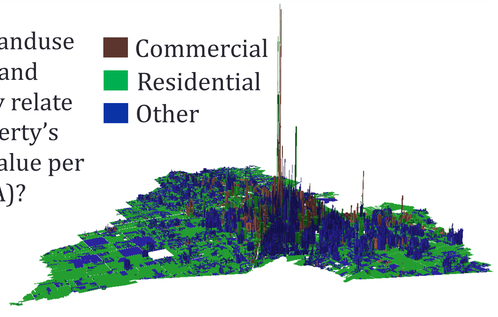
Mixed Use Downtown of San Miguel, Mexico →

The monotony of suburbia prompted me to ask:
what else is possible in urban design?

"...promote development patterns that secure a community's fiscal condition while reinforcing a stronger sense of place"
-Urban3 Mission Statement

RESEARCH QUESTION

How do landuse patterns and centrality relate to a property's market value per acre (VPA)?



METHODS

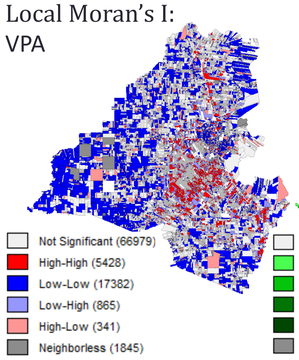
Spatial Autocorrelation?

Yes: Global Moran's I = 0.158

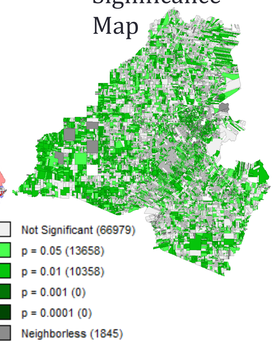
Significant?

Yes: p<0.001

Local Moran's I: VPA



Significance Map



Centrality Measures

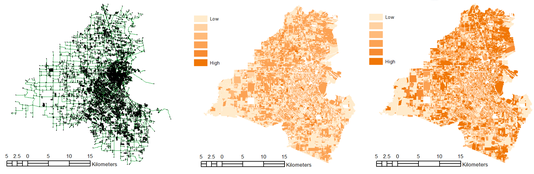
Distance to CBD: F= Di
Where Di is a parcel's Euclidean distance to the CBD.

Population Weighted Distance:
$P[i] = \sum_{i,j}(p)(b[i,j])$
Where $b[i, j]$ is the Euclidean distance between the centroids of blockgroups i and j;
p is the blockgroup population

Straightness: $S[i] = \frac{\delta[i,j]}{d[i,j]}$
Where $\delta[i, j]$ is the Euclidean distance between nodes i and j; $d[i,j]$ is the shortest network distance between the two nodes

Betweenness: $B[i] = \sum_{i,[j,k]}\left(\frac{Njk[i]}{Nji}\right)$
Where $Njk[i]$ = number of shortest paths from node j to node k that pass by node i, Njk = total number of shortest paths from node j to node k

Road Network **Betweenness** **Straightness**



Spatial Lag Hedonic Regression Model

$\ln(VPA) = \beta 0 + \beta 1 SpatialLag + \beta 2 DistanceToCBD + \beta 3 PopWeightedDistance + \beta 4 Straightness + \beta 5 Betweenness + \beta 6 Controls + e$

- Spatial Lag model chosen due to strong theoretical basis for spatial autocorrelation in value per acre
- Interactive terms for residential and other landuses were applied to all variables to control for commercial
- Rook-based contiguity chosen to minimize edge effects

RESULTS

(a) OLS with control variables only

Variable	(6) Log(VPA)	Statistical Significance
Constant	11.82004	***
Fragmentation	0.00162826	***
Residential*Fragmentation	0.00225094	***
Other*Fragmentation	0.00366743	***
% Alternative Transportation	1.120498	
Residential*% Alternative Transportation	-2.224976	***
Other*% Alternative Transportation	-3.839366	***
% Vacant	0.3129144	
Residential*% Vacant	-0.6344286	
Other*% Vacant	-1.176178	
% Owner occupancy	-0.800151	***
Residential*% Owner Occupancy	0.2867837	
Other*% Owner Occupancy	-0.8630236	***
% Black	-1.599148	***
Residential*% Black	0.7073772	***
Other*% Black	-1.007695	
Log (Per Capita Income)	0.09375673	*
Residential* Log (Per Capita Income)	1.34E-05	***
Other* Log (Per Capita Income)	1.25E-05	***
Residential	-2.248859	***
Other	-2.404345	***

$R^2 = 0.3699$

* p<0.05 ** P<0.01 *** p<0.001

(b) OLS with control variables and centrality measures

OLS Models	(1) Log(VPA)	(2) Log(VPA)	(3) Log(VPA)	(4) Log(VPA)	(5) Log(VPA)
Distance to CBD	-0.4200813 ***				-0.461528 ***
Residential*(Distance to CBD)	0.1821682 **				0.2494804 ***
Other *(Distance to CBD)	0.0107376				0.07343855
Population Weighted Distance		-0.279127 ***			-0.2025519 ***
Residential *Population Weighted Distance		0.2196147 ***			0.2292972 ***
Other *Population Weighted Distance		0.023528			0.06537528
Betweenness			0.02549 *		-0.00618767
Residential* Betweenness			-0.111307 ***		-0.07963821 ***
Other * Betweenness			-0.07425 ***		-0.07650639 ***
Straightness				-0.0166021	0.2548297 **
Residential* Straightness				-1.322589 *	-3.064056 ***
Other* Straightness				-0.6429154	-1.7821308 **

$R^2 = 0.4359$

Multi Collinearity?

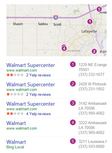
	Distance to CBD	Population Weighted Distance	Betweenness	Straightness
Distance to CBD	1			
Population Weighted Distance	-0.0339	1		
Betweenness	-0.0278	-0.0491	1	
Straightness	0.2111	0.0173	0.1413	1

Note: All centrality variables are divided by one standard deviation from the population mean.

Spatial Lag or Spatial Error Model?

TEST	MI/DF	VALUE	PROB
Lagrange Multiplier (lag)	1	613.1574	0.00000
Robust LM (lag)	1	451.0077	0.00000
Lagrange Multiplier (error)	1	531.7530	0.00000
Robust LM (error)	1	369.6033	0.00000

Spatial Lag Models	(1) Log(VPA)	(2) Log(VPA)	(3) Log(VPA)	(4) Log(VPA)	(5) Log(VPA)
Distance to CBD	-0.4092343 ***				-0.4587218 ***
Residential*(Distance to CBD)	0.185035 ***				0.2550478 ***
Other *(Distance to CBD)	0.01677907				0.08750631
Population Weighted Distance		-0.2873818 ***			-0.2135682 ***
Residential *Population Weighted Distance		0.2308836 ***			0.2382116 ***
Other *Population Weighted Distance		0.0272904			0.2382116
Betweenness			0.02561 *		-0.0067005
Residential* Betweenness			-0.1043401 ***		-0.07199643 ***
Other * Betweenness			-0.07104 ***		-0.06188681 ***
Straightness				-0.0166021	0.2803453 ***
Residential* Straightness				-1.322589 *	-3.118742 ***
Other*Straightness				-0.6429154	-1.879427 **
	$R^2 = 0.3677$	$R^2 = 0.368$	$R^2 = 0.356$	$R^2 = 0.36091$	$R^2 = 0.3797$



CONCLUSIONS

- High value of centrally located commercial properties suggests that several market failures may distort the urban growth process in Lafayette
- One limitation in this study relates to the lack of clear interpretation of the centrality coefficient magnitudes: only analyzing the signs and relative magnitudes of coefficients limits the ability of results to inform urban planning policy.
- More qualitative research could improve understanding of how different centrality measures impact value per acre to inform community design policies.